The University of Lahore
Department of CS & IT

Assignment 01

Instructor Name: Miss Iqra Adnan **Course Code: CS09202 | 11**
Course Title: Discrete Structures

Date: 21-10-2024 Deadline: 28-10-2024

PART I

Q1: Which of these are propositions? What are the truth values of those that are propositions?

a) Do not pass go.

b) What time is it?

c) There are no black flies in Maine

d) 4+x=5.

e) The moon is made of green cheese.

f) $2^n \geq 100$.

Q2: Let p, q, and r be the propositions

p :You have the flu.

q :You miss the final examination.

r :You pass the course.

Express each of these propositions as an English sentence.

a) $p \rightarrow q$

b) $q \rightarrow \neg r$

c) $\neg q \leftrightarrow r$

d) $p \vee q \vee r$

e) $(p \rightarrow \neg r) \vee (q \rightarrow \neg r)$

f) $(p \wedge q) \vee (\neg q \wedge r)$

Q3: Determine whether each of these conditional statements is true or false.

a) If 1+1 =3, then unicorns exist.

b) If 1+1 =3, then dogs can fly.

c) If 1+1 =2, then dogs can fly.

d) If 2 +2 =4, then 1+2 = 3.

Q4: Construct a truth table for each of these compound propositions.

a) $(p \leftrightarrow q) \oplus (p \leftrightarrow \neg q)$

b) $(p \oplus q) \rightarrow (p \oplus \neg q)$

c) $(\neg p \leftrightarrow \neg q) \leftrightarrow (p \leftrightarrow q)$

d) $(p \leftrightarrow q) \leftrightarrow (r \leftrightarrow s)$

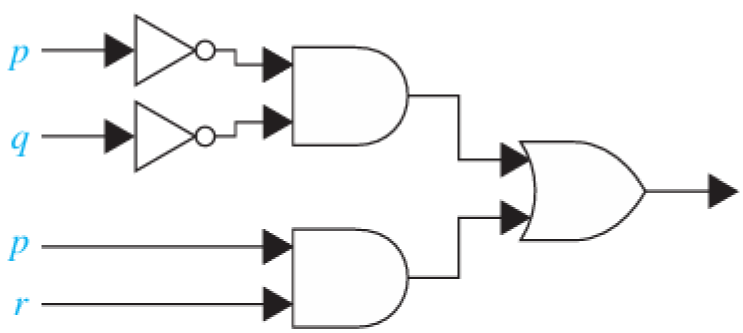
e) $(p \vee q) \wedge \neg r$

PART II

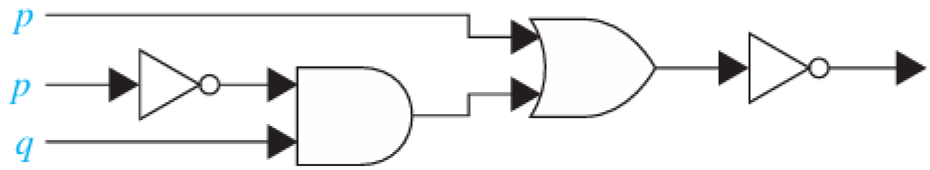
Q1: Construct a combinatorial circuit using inverters, OR gates, and AND gates that produces the output $(p \wedge \neg r) \vee (\neg q \wedge r)$ from input bits p, q, and r.

Q2: Construct a combinatorial circuit using inverters, OR gates, and AND gates that produces the output $((\neg p \vee \neg r) \wedge \neg q) \vee (\neg p \wedge (q \vee r))$ from input bits p, q, and r

Q3: Find the output of each of these combinatorial circuits if p=1,q=1, and r=1 .Also generate logical expression for the circuit.



Q4: Find the output of each of these combinatorial circuits if p=1,q=0, and r=1 .Also generate logical expression for the circuit.



Q5: Translate the English sentence into a logical expression?

"You can access the Internet from campus only if you are a computer science major or you are not a freshman."

PART III

Q1: Show that ¬(p →q) and p∧¬q are logically equivalent by developing a series of logical equivalences

Q2: Show that ¬(p →q) and p∧¬q are logically equivalent by developing a series of logical equivalences

Q3: Determine whether (¬p ∧ (p → q))→¬q is a tautology by developing a series of logical equivalences

Q4: Use truth tables to verify the absorption laws.

a) p∨(p∧q)≡p

b) p∧(p∨q)≡p

PART IV

Q1: Show that ∃x(P(x)∨ Q(x)) and ∃xP(x) ∨∃xQ(x) are logically equivalent.

Q2: Find a counterexample, if possible, to these universally quantified statements, where the domain for all variables consists of all real numbers. a) ∀x(x2= x) c) ∀x(|x| > 0)

Q3: Translate the statement ∃x∀y∀z((F(x,y) ∧ F(x,z)∧ (y= z)) →¬F(y,z)) into English, where F(a,b)means a and b are friends and the domain for x, y, and z consists of all students in your school.

Q4: Find a counterexample, if possible, to this universally quantified statement, where the domain for all variables consists of all integers.

a) ∀x∃y(x = 1/y)

PART V

Q1: Identify the error or errors in this argument that supposedly shows that if ∃xP(x) ∧∃xQ(x) is true then ∃x(P(x)∧Q(x)) is true.

1. ∃xP(x) ∨∃xQ(x) Premise

2. ∃xP(x) Simplification from (1)

3. P(c)

4. ∃xQ(x)

5. $Q(c)$

6. $P(c) \wedge Q(c)$ Existential instantiation from (2) Simplification from (1) Existential instantiation from (4) Conjunction from (3) and (5)

7. $\exists x(P(x) \wedge Q(x))$ Existential generalization

Q2: Use rules of inference to show that if $\forall x(P(x) \vee Q(x))$ and $\forall x((\neg P(x) \wedge Q(x)) \rightarrow R(x))$ are true, then $\forall x(\neg R(x) \rightarrow P(x))$is also true, where the domains of all quantifiers are the same.